

Experimental Analysis of Energy Transmission on Spring Coupled Pendula

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Abstract

This study investigates the dynamics of energy transmission between two spring-coupled pendula. The experiment was conducted using three distinct configurations: in-phase, out-of-phase, and single-stationary oscillations. The system parameters were experimentally determined and compared to a theoretical model derived from the linearized coupled oscillator equations. Data were collected via motion sensors and analyzed using Fast Fourier Transform (FFT) and exponential damping correction to isolate intrinsic oscillatory behavior. The results exhibit strong agreement between experimental and simulated data, confirming the presence of beat phenomena and predictable exponential energy decay. Moreover, the coupling distance s was found to directly influence the rate of energy transfer, validating the torque-dependent coupling model. These findings provide quantitative insight into the behavior of coupled oscillatory systems and energy exchange in weakly dissipative environments.

Keywords: Coupled pendulum, energy conservation, energy decay

1 Introduction

In classical mechanics, the simple pendulum serves as a foundational system for studying the conservation of mechanical energy and the interchange between kinetic and potential energy [1]. The motion of a single pendulum of length L and mass m is governed by the nonlinear second-order differential equation

$$mL^2\ddot{\theta} + mgL \sin \theta = 0, \quad (1)$$

where θ is the angular displacement from equilibrium. The total mechanical energy of the pendulum,

$$E = \frac{1}{2}mL^2\dot{\theta}^2 + mgL(1 - \cos \theta), \quad (2)$$

remains constant in the absence of nonconservative forces.

However, when two pendula are *mechanically coupled*, for instance, by a spring attached a distance s from each pivot and separated by a rest distance d , the system becomes dynamically rich, exhibiting mode coupling and energy transfer between the oscillators. The potential energy of the spring is given by

$$E_{\text{spring}} = \frac{1}{2}k(\ell - d)^2, \quad (3)$$

where ℓ is the instantaneous distance between spring attachment points. The generalized coordinates θ_1 and θ_2 now obey coupled nonlinear equations of motion,

$$\begin{aligned} mL^2\ddot{\theta}_1 + bL^2\dot{\theta}_1 + mgL \sin \theta_1 &= \tau_1(\theta_1, \theta_2), \\ mL^2\ddot{\theta}_2 + bL^2\dot{\theta}_2 + mgL \sin \theta_2 &= \tau_2(\theta_1, \theta_2), \end{aligned} \quad (4)$$

where b is some damping term, and τ_i are the torques induced by the spring coupling. These torques depend nonlinearly on θ_1, θ_2 and are derived from the geometry of the spring system [2]. In certain cases, we can simplify this by attaching the endpoints of the spring at a constant distance s from the pivot of each pendulum. For small oscillations ($\theta \ll 1$), the system reduces to a pair of linearly coupled harmonic oscillators (because $\sin x \approx x, x \ll 1$):

$$\ddot{\theta}_1 + \omega_0^2\theta_1 + \kappa(\theta_1 - \theta_2) = 0, \quad \ddot{\theta}_2 + \omega_0^2\theta_2 + \kappa(\theta_2 - \theta_1) = 0, \quad (5)$$

where $\omega_0 = \sqrt{g/L}$ and $\kappa = ks^2/(mL^2)$ is the coupling constant. The solutions exhibit normal modes, the symmetric (in-phase) and antisymmetric (out-of-phase) modes, with corresponding frequencies $\omega_{\pm} =$

$\sqrt{\omega_0^2 \pm \kappa}$. The superposition of these modes gives rise to *beats* [3], where energy oscillates periodically between the two pendula.

For an idealized closed system, the total energy remains conserved:

$$E_{\text{total}} = \sum_{i=1}^2 \left(\frac{1}{2} m L^2 \dot{\theta}_i^2 + m g L (1 - \cos \theta_i) \right) + E_{\text{spring}}. \quad (6)$$

In practice, damping from air resistance and pivot friction, among other energy losses, introduces dissipative losses that gradually decrease the system's total energy, though the relative energy exchange between the pendula remains observable over multiple cycles, and the energy loss is predictable enough to be factored out [4].

This experiment aims to quantitatively (qualitatively, if not able) analyze the transfer of energy in a spring-coupled pendulum system through experimental observation and numerical modeling. By initiating the pendula under different phase conditions, such as in-phase and out-of-phase configurations, we can compare measured energy exchange rates and beat frequencies with theoretical predictions and simulations, thereby characterizing the influence of coupling strength and damping on the dynamics of energy transmission.

2 Methodology

For the experimental setup as seen in Figure 1, two metal blocks with mass $M = (4.00 \pm 0.01)$ kg were each attached to metal rods with length $L = (115 \pm 1)$ cm, separated by distance $d = 0.20$ m from their pivots. Then, motion sensors (Vernier Motion Detector 2 [5]) were placed on the side of each metal block. A spring was used to connect the two metal bars to each other located a distance s from the top of the bar.

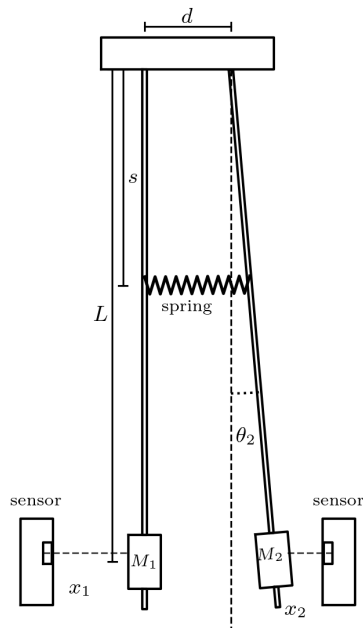


Figure 1: Diagram of the experimental setup

The two sensors can only measure the horizontal displacement x of each pendulum. In order to get θ , we must convert to $\theta = \sin^{-1} \left(\frac{x}{L} \right)$. Initially, two trials were performed in order to measure the spring constant k . Additionally, the pendulum parameters were also measured, such as the L, d, s , and so on. A mass was suspended in the spring and it was allowed to oscillate to determine the frequency of oscillation of the spring. From this, we can derive the k value. Next, we perform the oscillation trials. There are mainly three modes of performing trials, the pendulums are in-phase, out-of phase, and one is at rest. Each trial involved displacing one or both of the pendulums at a small angle ($\approx 5^\circ$) and the motion sensors recorded their oscillations. Additionally, "controls", i.e., springless uncoupled systems, were also recorded. The duration of the oscillations were recorded for 60 seconds for most of the trials (shorter durations were made in some when able). The process was repeated for various s , and various phase moods. We performed the trials with s of (10.0 ± 0.5) cm, (51.5 ± 0.5) cm, and (110.0 ± 0.5) cm. These distances aim to show varying torques of the coupled pendulum. Overall, there were 36 trials for each pendulum configuration. All the trials performed can be seen at the appendices.

The oscillation data was processed using Fast Fourier Transform (FFT) analysis to determine the relevant frequencies. The decay constant $e^{-b/2m}$ was determined and removed in order to minimize damping and accurately analyze the energy component. The energy of the system is analyzed, as well as its motion over time. Additionally, numerical simulations were performed in order to compare the experimental results with the theoretical calculations.

3 Results and Discussion

The k value of the spring was determined from the oscillation frequency of $f = 2.167 \text{ Hz}$ at $m = 0.1 \text{ kg}$. $k = m(2\pi f)^2 = 18.53 \text{ N/m}$. Additionally, from the period of the pendulum over multiple trials, it can be easily calculated that the center of mass $L = 114 \text{ cm}$, with high precision.

For the idealized simulations, we implement an algorithm known as velocity verlet integration, which is a symplectic integrator. We use the parameter values measured in the experiment proper, find an arbitrary value for b , and run the simulations. These are presented by Figure 2. Due to paper size constraints and page limits, we are not able to aesthetically show these graphs in the desired size.

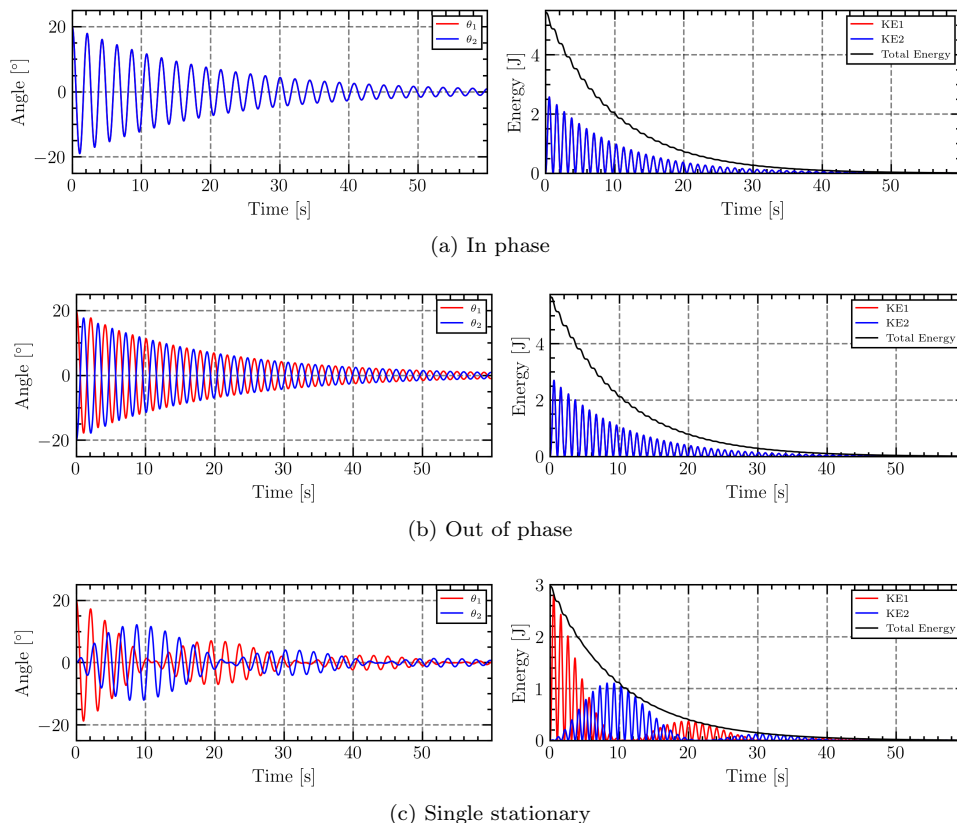


Figure 2: Spring coupled pendulum simulations showing both the position over time and the energy over time (dissipative). Parameters are $d = 20 \text{ cm}$, $s = 50 \text{ cm}$, and $L = 114 \text{ cm}$, exactly like the experimental parameters. Notice that in the two former modes, the energy is relatively stable and has an exponential decay rate. However, at Fig. 2c, we show an amplitude oscillation between the two, while the total energy remains the same (relatively, without the dampening) and continues to decay exponentially. These oscillations can be modelled by the normal modes $\omega = \sqrt{\omega_0^2 \pm \kappa}$.

The next figures show the experimental results. Due to the sheer number of trials done, the researchers will not be able to present all the data, and only a select sample of it will be shown. However, all the data and graphs is available in the provided GitHub repository in the appendices.

Figure 3 shows the time evolution of the pendula's angular displacements. The experimental results closely match the simulations, apart from minor stochastic noise. In both in-phase and anti-phase modes, the total energy approximates twice the kinetic energy of either pendulum, consistent with theory. The beat frequency, arising from the normal mode superposition, can be extracted via FFT but is not analyzed here. Other data (not shown here, but evident in the appendices) shows that Energy transfer is strongly influenced by the spring attachment distance s . Larger s values yield stronger coupling and faster exchange due to increased torque from a longer moment arm.

The characterization of energy decay within the coupled system presents a particular challenge in data analysis, as the total energy dissipates over sufficiently long timescales, limiting the observable dynamics. Nevertheless, this decay follows a predictable exponential form, $E(t)^{-\alpha t}$ where $\alpha \in \mathbb{R}^+$. To mitigate this damping effect, one may effectively “remove” the exponential decay by multiplying the

system response by the inverse factor $e^{\alpha t}$. Practically, this is achieved by fitting the observed energy curve to an exponential decay model and applying the corresponding compensating factor to the dataset. The resulting transformation yields a pseudo-energy-conserving system, allowing for further analysis with improved signal retention. Figure 4 illustrates this correction, where the apparent decay is eliminated, thereby enhancing the visibility of intrinsic dynamical features. It should be noted, however, that the method is sensitive to parameter estimation; inaccuracies in the fitted decay constant can lead to artificial divergence or amplification at extended timescales.

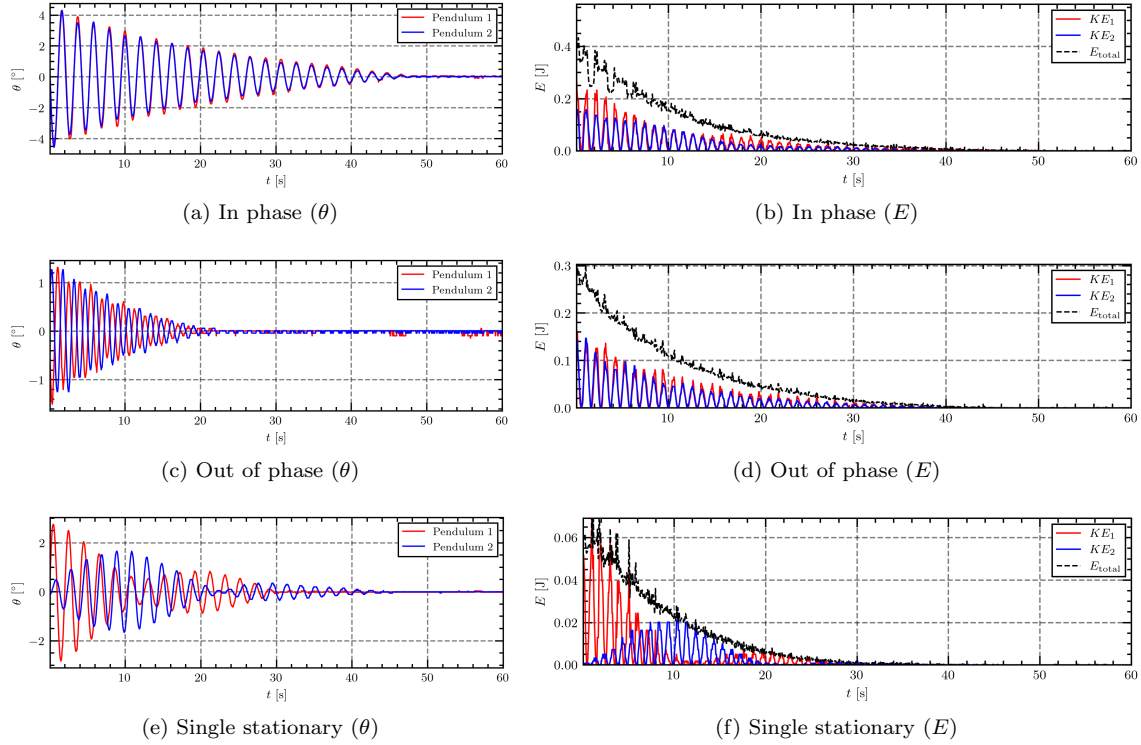


Figure 3: Angular displacement and energy profiles for $s = 50$ cm are shown, with the same units as Figure 2. The experimental data align closely with simulations, deviating only by minor noise. Energy decays exponentially while remaining locally conserved, and Figs. 3e-3f clearly demonstrate the predicted energy transfer between pendula, with the total energy remaining constant but KE being shared among the two

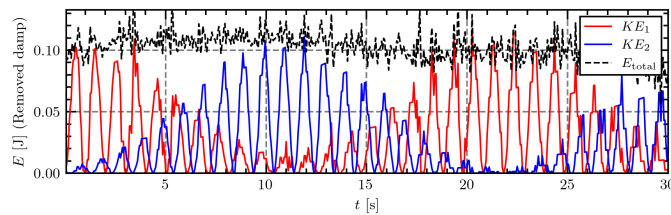


Figure 4: Fig 3f, with the energy decay removed, resulting in the mean energy to be relatively constant, albeit noisy. Notice how we see almost perfect sinusoidal energy transfers, making the system pseudo-conservative

4 Conclusion

The experiment successfully verified the transfer of energy between two spring-coupled pendulums, consistent with the numerical simulations of coupled oscillators. The in-phase and out-of-phase motions that were observed matched the expected normal modes. Additionally, for single stationary setups, the beat phenomenon was observed. We determined that energy transfer was dependent on the coupling strength which increases directly proportional to the spring attachment distance s . Although energy decay was present, we were able to effectively "remove" the exponential decay to achieve a pseudo-energy-conserving system. Overall, the results showed that the coupled pendulum system exhibited the predicted energy transfer based on the theoretical model of linearly coupled oscillators.

References

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Appendices

Table 1: Table of Contributors

Member	Contribution
Danielle Louis Dumas	Performed the experiment, wrote the Methodology and Conclusion
Vercil Juan	Performed the experiment, wrote the Results and Discussion, performed Data Analysis
Rwen Josh Narvarte	Performed the experiment, wrote the Introduction and Abstract

Table 2: Summary of Coupled Pendulum Trials

Run Type	Trials	Duration [s]	s [cm]	Notes
In-phase	1–5	5	51.5 ± 0.5	Initial short-duration trials
Out-of-phase	6–10	5	51.5 ± 0.5	Initial short-duration trials
Left/Right at rest	11–14	45–60	51.5 ± 0.5	Observation of energy transfer
In-phase	15, 19–20, 25–26	60	10.0 ± 0.5	Full-length runs
Out-of-phase	16, 21–22, 27–28	60	10.0 ± 0.5	Full-length runs
Left/Right at rest	23–24, 29–32	60	110.0 ± 0.5	Tests of energy transfer
Control (no spring)	35–36	60	–	Baseline for uncoupled motion
k -measuring	17–18	30	–	Determination of coupling constant

Documentation

Acknowledgments

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Code

All the code used for the graphing and statistical analysis for this report can be found in:

<https://github.com/silvernuke911/Physics157/tree/main/PHYSICS192%20CODE/E1B>